Echolocation Mapping System

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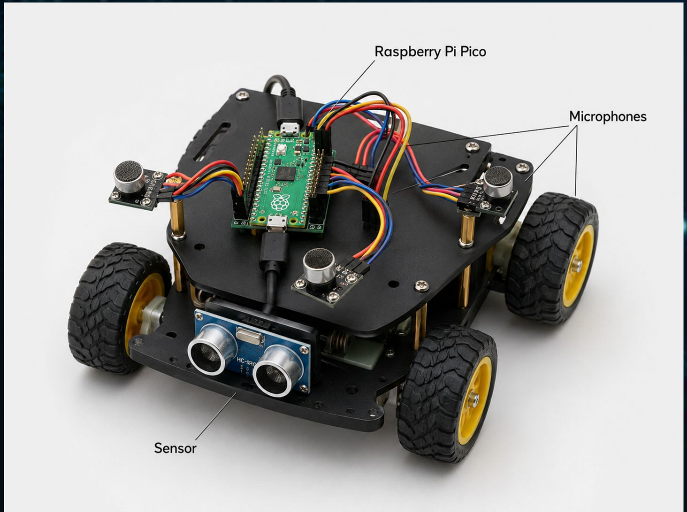
Project Overview

Purpose: Exploration of underground systems such as caves, collapsed mines, pipelines, etc.

Theory: Mimic echolocation phenomena used by bats and dolphins, and translate the analog frequency responses into digital representation.



Project Overview

Challenges:

- Implementation of environmental factors in a MATLAB simulation
- Prevent signal attenuation
- Calculation of precise values.

Approach & Deliverables

Level 1

- → Signal Generation
- → Echo Signal Modeling
- → Simulation Generation

Level 2

- → Convolution Calculations
- → Digital Filtering
- → Fast Fourier Transform Analysis

Level 3

- → Two-Dimensional Mapping
- → Material Classification
- → Error Refining

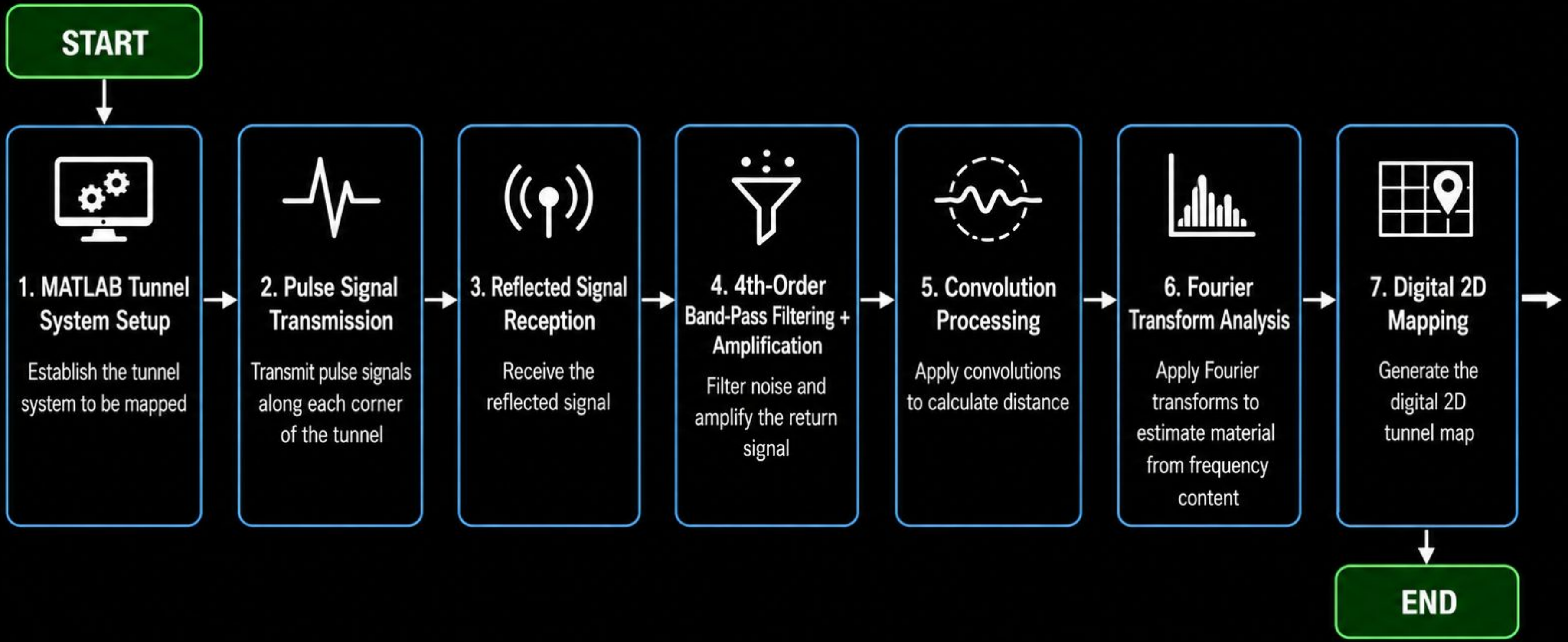
Methods

To develop the MATLAB simulation for the echolocation-based mapping system, we had to focus on the following subgroups to create a realistic software to accommodate for the factors we lose by not implementing hardware in our design.

Methods include:

- Signal Generation
- Tunnel System Environment
- Convolutions, Fourier Transforms
- Digital Filtering (FIR-BPF)
- Mapping System
- Error Factor Accomodation

TUNNEL MAPPING PROJECT – SOFTWARE FLOW CHART

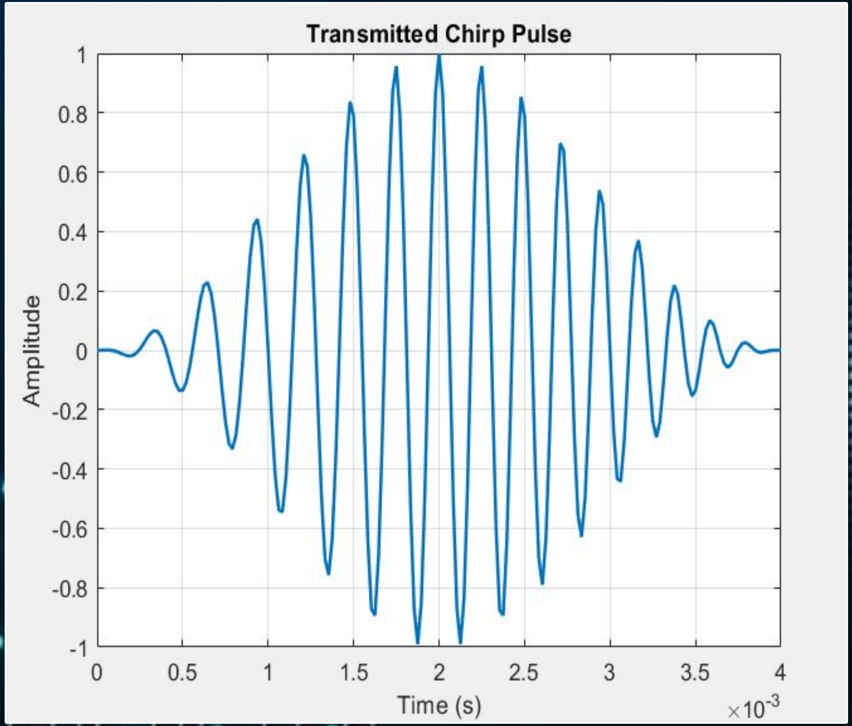


Signal Generation

What kind of acoustic signal is the most efficient for frequency content analysis?

Initial Approach: Opt for a simple impulse signal at around 4800 Hz.

Updated Approach: Use a frequency sweep pulse (chirp) from 3000 to 5000 Hz over 4 millisecond period.

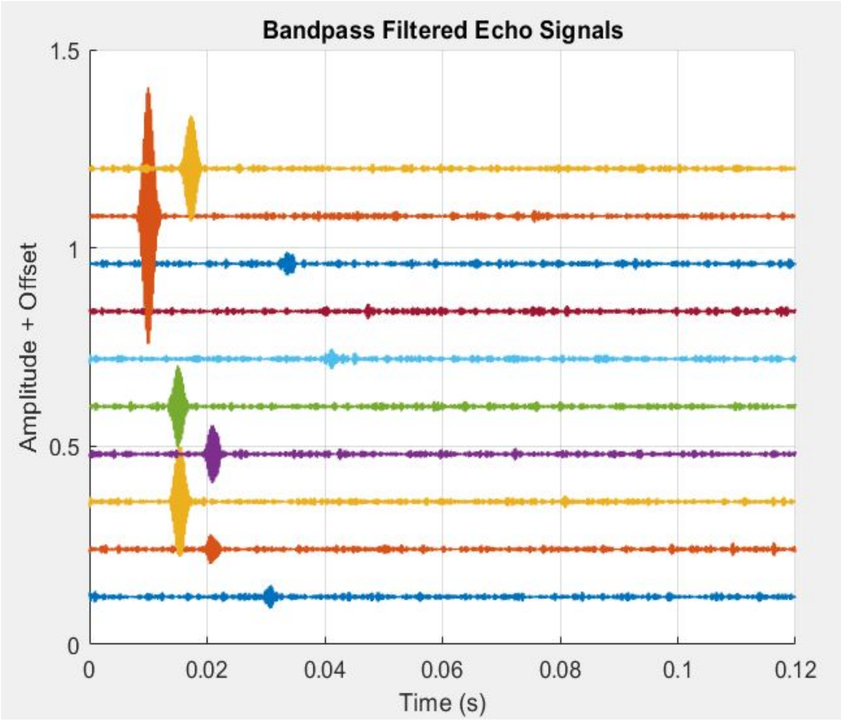


Finite Impulse Response Band-Pass Filtering

Implemented a digital butterworth band-pass filter function in MATLAB

Low-cut frequency of 3000 Hz and high-cut frequency of 5000 Hz.

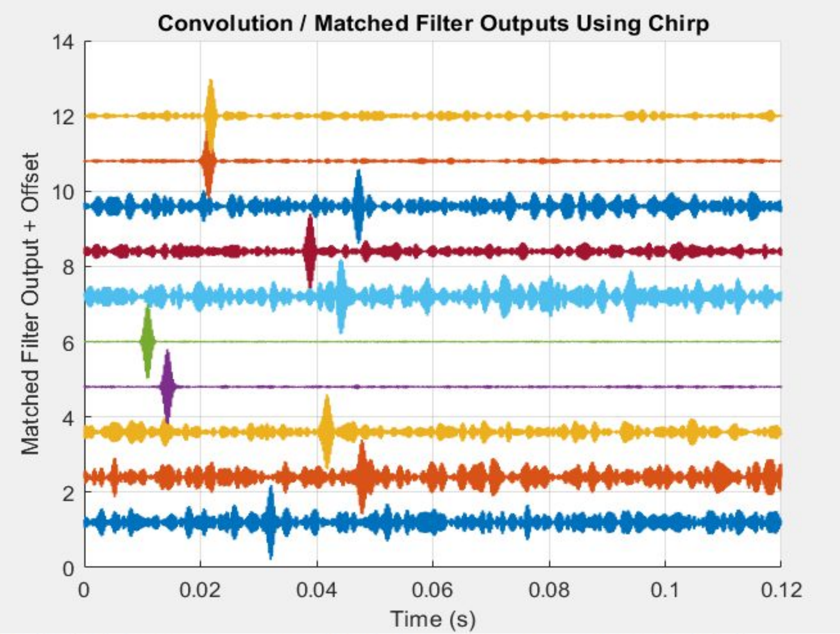
Chose a 4th-order Butterworth band-pass filter because it provides a sharper cut off than lower-order filters, allowing effective isolation of the 3–5 kHz chirp range.



Calculations using Convolution

Generate a random frequency chirp response in MATLAB

Convolve initial response and time reversed transfer function using basic formula [X(s)*H(-s) = Y(s)] at highest frequency time interval



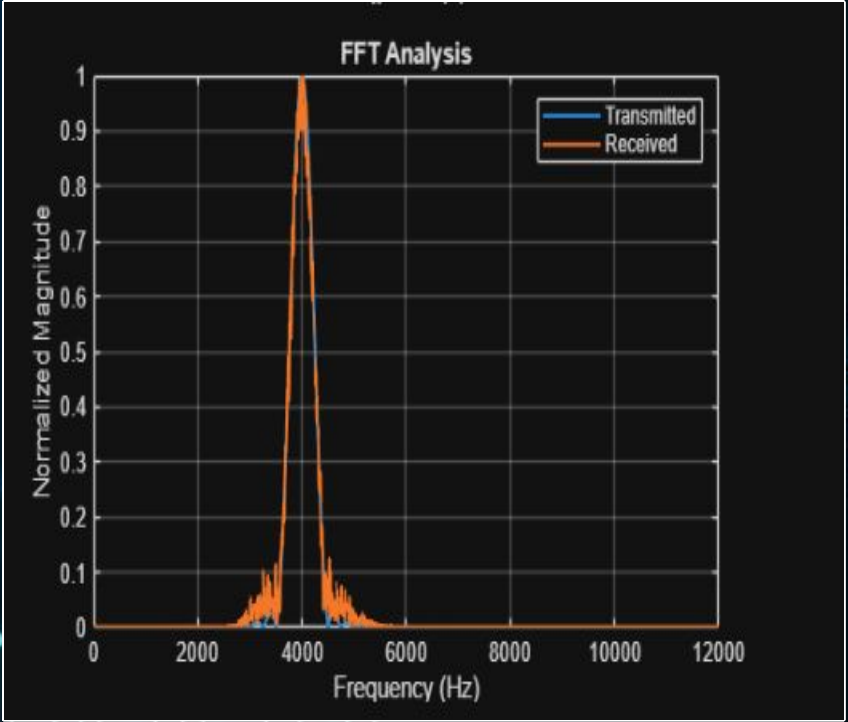
ActualDistance_m	RawEstimatedDistance_m	CorrectedDistance_m
1.2402	1.2398	1.2522
3.7708	3.7623	3.7247
3.9757	3.9838	3.944
2.3192	2.3117	2.2886
2.4506	2.4689	2.4442
9.7895	17.525	17.35

Fourier Transforms

FFT converts signal from time domain to frequency domain

Used to verify Frequency range, allowing for material estimation

Confirms bandpass filter preserved the desired echo frequency



Two-Dimensional Mapping Simulation

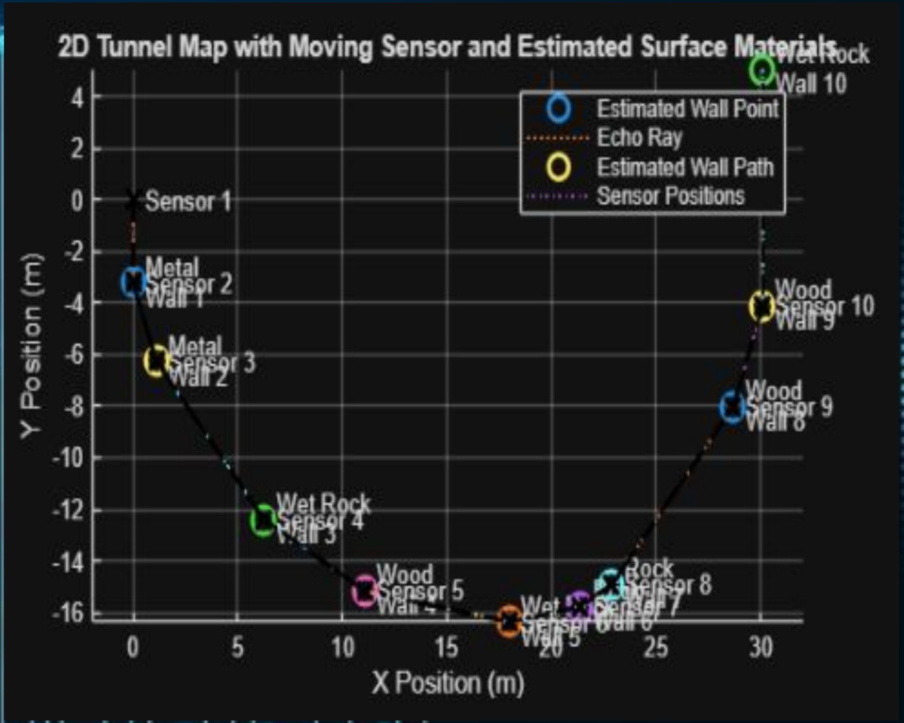
Distances converted to coordinates.

Multiple measurements create tunnel shape.

Sensor moves after each scan.

Echo rays show path from sensor to wall.

Material labels add environmental information.



Conclusions

Due to the time frame of this project and limited resources, hardware implementation was not within the scope of this project.

Given more time and resources, we would consider adding analog filtering, the speaker-microphone setup for signal processing, and the implementation of the autonomous vehicle that moves based on the factors calculated by reflected signals.

Learning Outcomes

To make life easier for this project, we would implement the following:

- → Real as opposed to randomly generated data through hardware implementation
- → 360 degree microphone setup to better estimate reflection angle
- → Use of match filtering to isolate a known frequency range for the best data collection
- → Experimentation of different frequencies and filter orders to best reduce noise